



HK IMO Training 2021-2022
Problem Set 1 Suggested Solutions



Problem 1 (Caucasus MO 2021 Seniors Problem 1). The integers from 1 to 100 are written in a row in some order. A number is *good* if it is greater than all numbers to the right of it. A number is *nice* if it is greater than all numbers to the left of it. Suppose there are exactly k good numbers and exactly k nice numbers. Find the largest possible value of k .

Solution. The largest possible value of k is 50.

If the number 100 is put in one of the first 50 positions (counted from the left), then all numbers in the last 50 positions are not nice since they are smaller than 100. Similarly, if the number 100 is put in one of the last 50 positions, then all numbers in the first 50 positions are not good. Therefore, we have $k \leq 50$ in any case.

The largest possible value $k = 50$ can be attained when the numbers are $1, 2, \dots, 49, 100, 50, 99, 98, \dots, 51$, for example. Note that $1, 2, \dots, 49, 100$ are nice, while $100, 99, \dots, 51$ are good. □

Problem 2 (Dutch TST 2015 Day 1 Problem 5). For each positive integer n , let d_n be the largest integer that divides $a^n + (a + 1)^n + (a + 2)^n$ for every positive integer a .

(a) Prove that for every positive integer n , the number d_n is of the form 3^k for some nonnegative integer k .

(b) Prove that for every nonnegative integer k , there exists a positive integer n such that $d_n = 3^k$.

Solution.

(a) Consider any prime p dividing d_n . Note that $p \mid d_n \mid p^n + (p + 1)^n + (p + 2)^n$ and $p \mid d_n \mid (p + 1)^n + (p + 2)^n + (p + 3)^n$. Taking their difference, we obtain $p \mid (p + 3)^n - p^n$. This yields $p \mid 3^n$, and so p can only be 3. This shows d_n can only be a power of 3.

(b) For $k = 0$, we claim that $d_2 = 1 = 3^0$. Indeed, note that $d_2 \mid 1^2 + 2^2 + 3^2 = 14$. Since d_2 is a power of 3, we must have $d_2 = 1$.

For $k \geq 1$, we claim that $d_n = 3^k$ when $n = 3^{k-1}$. Firstly, for any positive integer a , the numbers $a, a + 1, a + 2$ are congruent to $0, 1, 2$ modulo 3 in some order. Suppose $\{a, a + 1, a + 2\} = \{3x, 3y + 1, 3z + 2\}$ where $x, y, z \in \mathbb{Z}$. By the lifting the exponent lemma, we have

$$v_3((3y + 1)^n + (3z + 2)^n) = v_3((3y + 1) + (3z + 2)) + v_3(n) = v_3(3(y + z + 1)) + v_3(3^{k-1}) \geq 1 + (k - 1) = k.$$

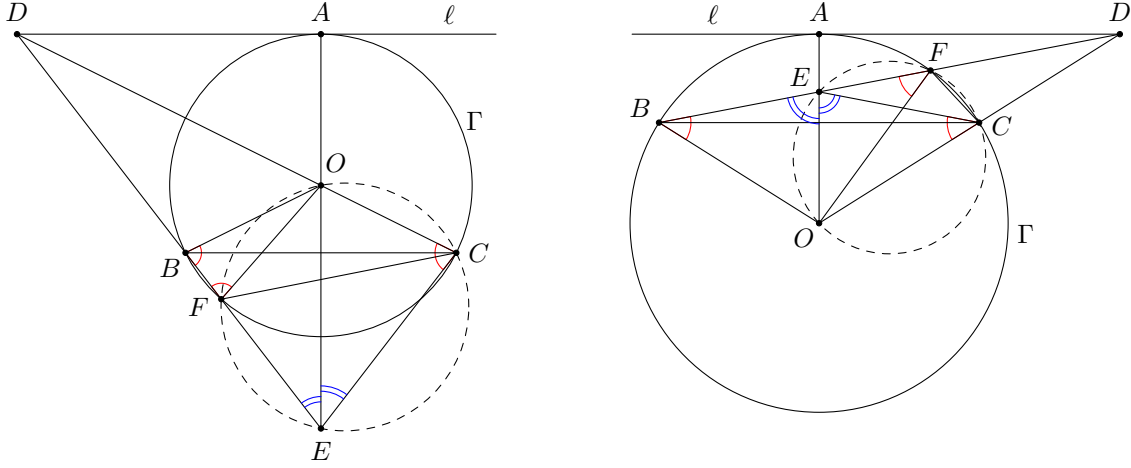
Also, we have $v_3((3x)^n) \geq n = 3^{k-1} \geq k$ (by induction). This implies $v_3(a^n + (a + 1)^n + (a + 2)^n) \geq k$. This shows $3^k \mid a^n + (a + 1)^n + (a + 2)^n$ for all $a \in \mathbb{N}$.

Next, we show that $v_3(1^n + 2^n + 3^n) = k$. If this holds, then $3^{k+1} \nmid 1^n + 2^n + 3^n$, and so $d_n = 3^k$. Indeed, this clearly holds when $k = 1$ (and $n = 1$). When $k \geq 2$, we have

$$v_3(1^n + 2^n) = v_3(1 + 2) + v_3(n) = 1 + (k - 1) = k$$

and $v_3(3^n) = n = 3^{k-1} \geq k + 1$ (again by induction). Therefore, $v_3(1^n + 2^n + 3^n) = k$, and we are done. □

Problem 3 (Slovenia TST 2018 Problem 4). Let O be the centre of a circle Γ . Let A be a point on Γ with tangent ℓ . A line parallel to ℓ cuts Γ in two points B and C . The line CO intersects ℓ at D . Let E be the intersection point of the lines AO and DB . Prove that the line passing through C and parallel to the tangent at O to the circumcircle of $\triangle OCE$ intersects the line BD on Γ .



Solution. Let BD meet Γ again at F . Since BC is parallel to the tangent at A , the line OA is the perpendicular bisector of BC . Therefore, we have $\angle OCE = \angle EBO$. Also, as $OB = OF$, we have $\angle FBO = \angle OFB$. This implies

$$\angle OCE = \angle EBO = \angle FBO = \angle OFB = \angle OFE,$$

and so O, C, E, F are concyclic. Now, since $OC = OF$, the tangent at O to (OCF) is parallel to CF . This proves the result. $\square$

Problem 4 (USA TSTST 2014 Problem 3). Find all polynomials $P(x)$ with real coefficients such that

$$P(\sqrt{2}x) = P(x + \sqrt{1 - x^2})$$

for all real number x with $|x| \leq 1$.

Solution. $P(x)$ can be any polynomial of $T_8\left(\frac{x}{\sqrt{2}}\right)$, where T_8 is the 8th degree Chebyshev polynomial of the first kind (see remark). Let $x = \cos \theta$ for some $\theta \in [0, \pi]$. Then the condition becomes

$$P(\sqrt{2} \cos \theta) = P(\cos \theta + \sin \theta) = P\left(\sqrt{2} \cos\left(\theta - \frac{\pi}{4}\right)\right).$$

Define $Q(x) = P(\sqrt{2}x)$. Then this further becomes

$$Q(\cos \theta) = Q\left(\cos\left(\theta - \frac{\pi}{4}\right)\right). \quad (1)$$

For $\theta \in [\pi, 2\pi]$, since $2\pi - \theta \in [0, \pi]$, we also have

$$Q(\cos \theta) = Q(\cos(2\pi - \theta)) = Q\left(\cos\left(2\pi - \theta - \frac{\pi}{4}\right)\right) = Q\left(\cos\left(\theta + \frac{\pi}{4}\right)\right).$$

Combining these, it is not hard to see that (1) indeed holds for all θ . We want to show that Q is any polynomial of $T_8(x)$.

Firstly, if $Q(x) = f(T_8(x))$ for some $f \in \mathbb{R}[x]$, then we have

$$Q(\cos \theta) = f(T_8(\cos \theta)) = f(\cos(8\theta)) = f(\cos(8\theta - 2\pi)) = f\left(T_8\left(\cos\left(\theta - \frac{\pi}{4}\right)\right)\right) = Q\left(\cos\left(\theta - \frac{\pi}{4}\right)\right).$$

Secondly, suppose Q satisfies (1) for all θ . Clearly, we can write $Q(x) = f(T_8(x)) + g(x)$ for some $f, g \in \mathbb{R}[x]$ where $\deg g < 8$. By (1) and the above deduction, we easily obtain $g(\cos \theta) = g\left(\cos\left(\theta - \frac{\pi}{4}\right)\right)$. Now, we have

$$g(\cos 1) = g\left(\cos\left(1 - \frac{\pi}{4}\right)\right) = g\left(\cos\left(1 - \frac{2\pi}{4}\right)\right) = \dots = g\left(\cos\left(1 - \frac{7\pi}{4}\right)\right).$$

Since $\cos\left(1 - \frac{j\pi}{4}\right) \neq \cos\left(1 - \frac{k\pi}{4}\right)$ for distinct $j, k = 0, 1, \dots, 7$, this shows g has the same value at 8 different points. As $\deg g < 8$, the polynomial g must be a constant function. This shows Q is a polynomial of $T_8(x)$. So we are done. $\square$

Remark. For each positive integer n , the Chebyshev polynomial T_n of the first kind is a polynomial such that $T_n(\cos \theta) = \cos(n\theta)$ for all $\theta \in \mathbb{R}$. The Chebyshev polynomial U_n of the second kind is a polynomial such that $U_n(\cos \theta) \sin \theta = \sin((n+1)\theta)$ for all $\theta \in \mathbb{R}$. It can be shown that T_n and U_n are degree n polynomials.